

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (All Branches) Examination
May 2011

CL103 Mechanics of Solids

Date: 18.05.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Find center of gravity of an I section having top flange 240mm x 10mm, bottom flange 180mm x 10mm and web 300mm x 10mm. 07
- Q - 2 (a) Define Limiting Friction, Angle of Friction, Angle of Repose and Coefficient of Friction. What are laws of static friction? 07
- (b) Find the forces in all members of the truss shown in Fig. 1. 07

OR

- Q - 2 (a) A ladder 6m long rests on horizontal ground and leans against a smooth vertical wall at an angle of 20° with vertical. Its weight is 900 N acting at its middle. It is on the point of sliding when a man weighing 500 N stands on it at a distance 2m along the ladder from foot of the ladder. Calculate the coefficient of friction. 07
- (b) Find the forces in all members of the truss shown in Fig. 2. 07
- Q - 3 (a) State & prove law of parallelogram. 04
- (b) Find resultant of the force system shown in Fig. 3. 05
- (c) Find resultant of the force system shown in Fig. 4 with respect to point A. 05

OR

- Q - 3 (a) Explain principles of superposition & transmissibility. 04
- (b) The sum of two concurrent forces P and Q is 270 N and their resultant is 180 N. The angle between the force P and resultant R is 90° . Find the magnitude of each force and angle between them. 05
- (c) What is Moment? Differentiate between moment and couple. Give characteristics of a couple. 05

SECTION - II

- Q - 4 (a) Attempt any one: 03
- (i) Define Amplitude, Frequency and Natural time period.
- (ii) Derive equation of un-damped SDOF system.
- (b) Describe how you will measure the hardness of the given metal in the laboratory. 04
- Q - 5 (a) A stepped bar is loaded as shown in Fig. 5. Calculate the stresses in each part and total change in the length of the bar. Take $E_s = 200$ GPa, $E_c = 100$ GPa and $E_b = 80$ GPa. 07

OR

- Q - 5 (a) A load of 2000 kN is applied on a short concrete column 500mm x 500mm, 07

reinforced with four 10mm steel bars. Find stresses in concrete and steel. Take value of E for steel as $2.1 \times 10^5 \text{ N/mm}^2$ and for concrete $1.4 \times 10^5 \text{ N/mm}^2$.

Q - 5 (b) Define Poisson's ratio, Bulk modulus 02

Q - 5 (c) A bar 0.5m long is rectangular in section with width 40mm and thickness 30mm. Calculate the change in dimensions when a tensile load of 120 kN is acting along its longitudinal axis if $E = 200 \text{ kN/mm}^2$ and Poisson's ratio 0.25. 05

OR

Q - 5 (c) A steel rod of 16mm diameter and 5m length, is connected to two grips at each end at a temperature of 150°C . Find the pull exerted when the temperature falls to 40°C if the ends (i) Do not yield (ii) Yield by 1 mm. 05

Take coefficient of thermal expansion as $12 \times 10^{-6} \text{ per } ^\circ \text{C}$.

Q - 6 Draw Shear Force & Bending Moment diagrams for any two beams from figures Fig. 6(a), Fig. 6(b) and Fig. 6(c) indicating values at all important points. Find point of contra-flexure if any. 14

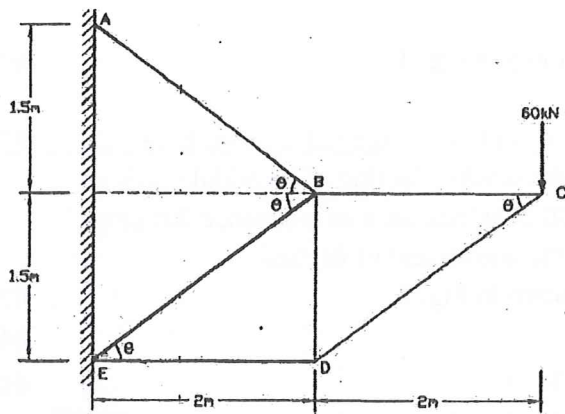


Fig. 1

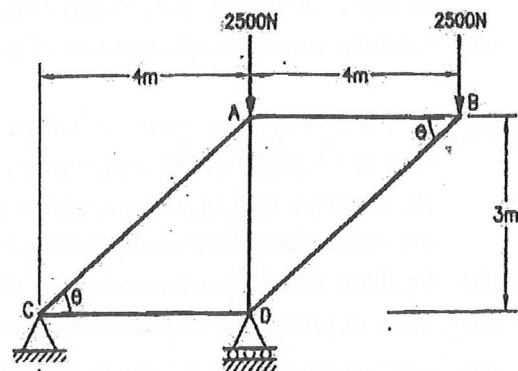


Fig. 2

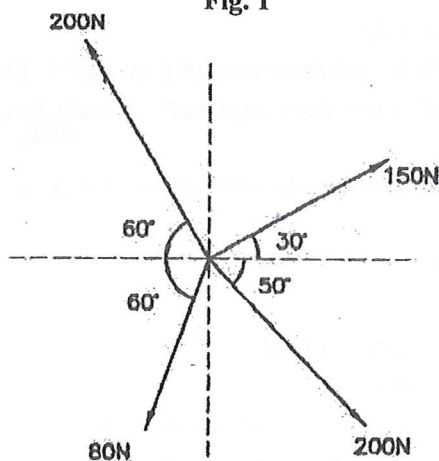


Fig. 3

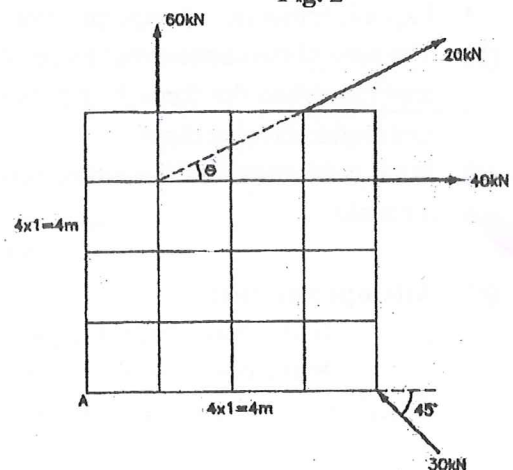


Fig. 4

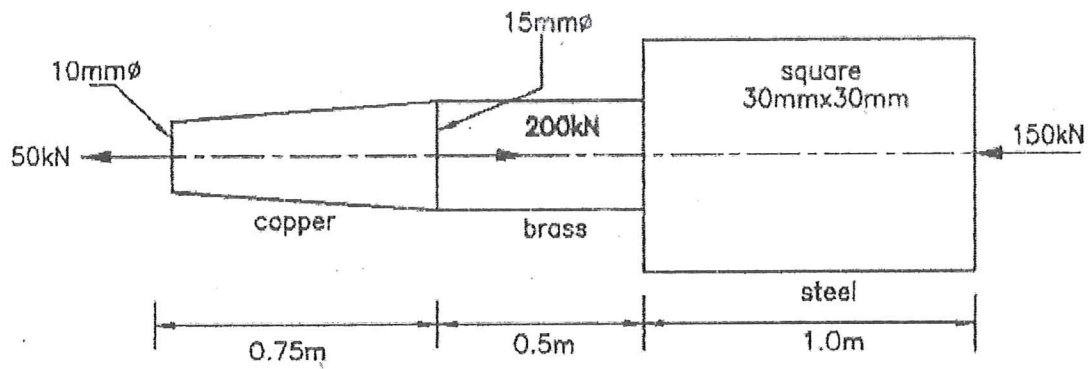


Fig. 5

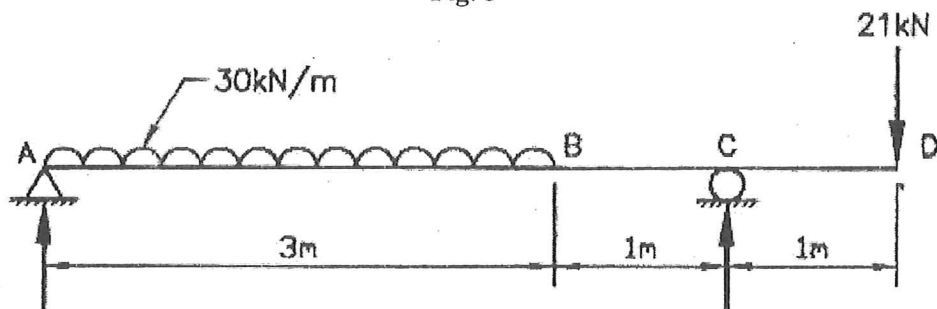


Fig. 6(a)

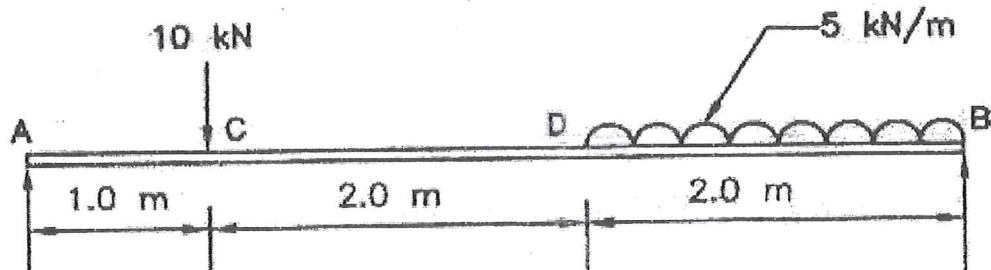


Fig. 6(b)

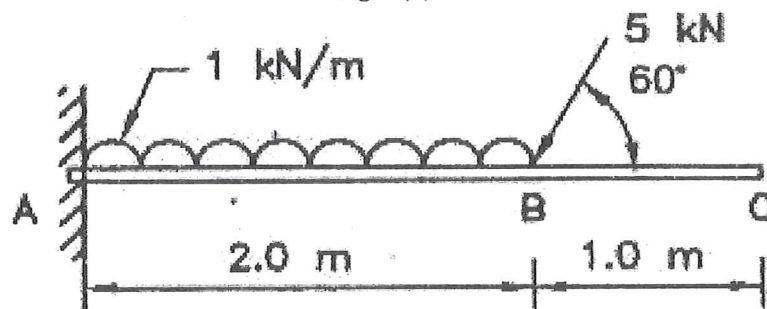


Fig. 6(c)
